BHANU SHANKAR

FULL STACK DEVELOPER | REACT | PYTHON | FASTAPI | AI/GENAI

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SUMMARY

Full Stack Developer with 6.5+ years of professional software development experience at Amzur Infotech Pvt. Ltd., with strong experience in modern frontend, backend, database, API, and AI application development. Experienced in building web applications using React, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, Vite, Python, FastAPI, REST APIs, SQLAlchemy, PostgreSQL, and Supabase. Hands-on experience developing Generative AI and LLM-powered applications using LangChain, LangGraph, Retrieval-Augmented Generation (RAG), ChromaDB, embeddings, Gemini, LiteLLM, AI agents, and multimodal AI. Experienced in authentication, authorization, database integration, API development, third-party integrations, testing, debugging, and AI-assisted software development using GitHub Copilot and Cursor. Experienced in Vibe Coding and prompt-driven development for rapid prototyping, code generation, refactoring, debugging, testing, and full-stack application development. Strong experience working across the complete software development lifecycle from requirements and development through testing, integration, debugging, and deployment.

WORK EXPERIENCE

Amzur Infotech Pvt. Ltd. | Visakhapatnam **Nov 2019 - Present**

FULL STACK DEVELOPER

Full Stack Development

- Develop full-stack web applications using React.js, Next.js, JavaScript, TypeScript, Python, FastAPI, Node.js, Express, REST APIs, PostgreSQL, and MySQL.
- Build reusable and responsive frontend applications using React, Next.js App Router, Vite, Tailwind CSS, React Router, TanStack Query, Zustand, Axios, Zod, and modern component libraries.
- Develop backend services and REST APIs using Python, FastAPI, Node.js, Express, Pydantic, SQLAlchemy, Alembic, Uvicorn, and Flask.
- Design and integrate relational databases using PostgreSQL, MySQL, Supabase, SQLAlchemy, asyncpg, and psycopg2.
- Implement authentication and authorization using JWT, HttpOnly cookies, bcrypt, Google OAuth 2.0, NextAuth, and Casbin RBAC.
- Develop API validation, error handling, CORS, input validation, file validation, rate limiting, audit logging, and security controls.
- Integrate external services including Google Calendar, Google Drive, Google Directory, Gmail, Cloudinary, Supabase, SMTP, and n8n.
- Work with MySQL stored procedures, triggers, views, background jobs, scheduled tasks, and database-driven business workflows.
- Develop automated tests using pytest, pytest-asyncio, pytest-cov, Vitest, React Testing Library, and HTTPX.
- Use Git and GitHub for version control, branching, pull requests, code reviews, and collaborative development.

AI / Generative AI Development

- Develop Generative AI and LLM-powered applications using LangChain, LangGraph, Gemini, LiteLLM, ChromaDB, embeddings, RAG, and AI agents.
- Build Retrieval-Augmented Generation pipelines covering document ingestion, chunking, embeddings, vector search, semantic retrieval, query rewriting, grounded generation, and source attribution.

- Develop AI agents and tool-based workflows for natural-language SQL querying, spreadsheet analysis, research, task detection, sentiment analysis, and conversational applications.
- Integrate multimodal AI capabilities for text, images, video, PDF, Excel, code, and structured data.
- Use LiteLLM as an LLM proxy and model abstraction layer for AI applications.

AI-Assisted Development & Vibe Coding

- Use GitHub Copilot and Cursor for AI-assisted software development, including code generation, refactoring, debugging, testing, and technical documentation.
- Apply Vibe Coding workflows to translate natural-language requirements into frontend, backend, API, database, and AI application features.
- Follow an iterative workflow of prompting, reviewing generated code, running tests, identifying issues, debugging, and refining implementations.
- Review and validate AI-generated code for correctness, security, maintainability, performance, and compatibility before integration.

TECHNICAL SKILLS

Frontend Development: React.js, React 18, React 19, Next.js 15/16, JavaScript ES6+, TypeScript, HTML5, CSS3, Tailwind CSS, Bootstrap, Vite, React Router, TanStack Query, Zustand, Axios, Zod, TanStack Table, shadcn/ui, Radix UI, Ant Design, PrimeReact

Backend Development: Python, FastAPI, Node.js, Express.js, Flask, REST APIs, Pydantic, SQLAlchemy 2.0, Alembic, Uvicorn, Gunicorn, Async Programming, API Development, API Integration, Webhooks, CORS

Databases: MySQL, PostgreSQL, Supabase PostgreSQL, SQL, SQLAlchemy, asyncpg, psycopg2, SQLite, ChromaDB, Stored Procedures, Triggers, Views

AI / Generative AI: Generative AI, LLM Applications, LangChain, LangGraph, RAG, Retrieval-Augmented Generation, AI Agents, Semantic Search, Vector Search, Embeddings, Prompt Engineering, Gemini 2.5 Flash, Gemini Imagen 4, LiteLLM, Multimodal AI, MCP

AI-Assisted Development & Vibe Coding: Vibe Coding, GitHub Copilot, Cursor, AI-Assisted Coding, AI Code Generation, Prompt-Driven Development, AI-Assisted Refactoring, AI Debugging, AI Test Generation, AI Documentation, Rapid Prototyping

Authentication & Security: JWT, HttpOnly Cookies, bcrypt, Google OAuth 2.0, NextAuth, Casbin, Role-Based Access Control, Input Validation, MIME Validation, API Security, Path Traversal Protection, Helmet, Rate Limiting

Integrations & Services: Google Workspace, Google Calendar, Google Drive, Google Directory, Gmail, Cloudinary, SMTP, Supabase, n8n, REST Integrations, Webhooks, Nodemailer

Testing & Quality: pytest, pytest-asyncio, pytest-cov, Vitest, React Testing Library, HTTPX, Unit Testing, Integration Testing, Code Review, Debugging

Tools & Practices: Git, GitHub, NPM, ESLint, Storybook, Swagger/OpenAPI, Vercel, Netlify, Agile, Scrum, Node Cron

PROJECTS

RevUp Lead Tracker — Full-Stack Sales CRM

Tech: Next.js 15/16, React 19, Tailwind CSS 4, Node.js, Express, FastAPI, Flask, MySQL, SQLAlchemy, JWT, Casbin, Google APIs, Gemini, Cloudinary

- Built a full-stack sales CRM with a Next.js client, separate Next.js admin panel, Express CRM API, FastAPI admin API, and MySQL stored procedures.
- Delivered lead, deal, contact, and company workflows with dashboards, sales funnel, calendar, Google Drive documents, CSV import, global search, and PDF/Excel export.
- Implemented admin authentication using JWT, HttpOnly sessions, bcrypt password handling, SMTP password reset, and Casbin role and business-unit permissions.
- Integrated Google Workspace services including OAuth, Google Calendar, Google Drive, Google Directory, and Gmail, plus Cloudinary for profile images.
- Added sentiment scoring using Flask, PyTorch, Hugging Face BERT/RoBERTa models, and Gemini for comment, lead, news, and company sentiment.
- Built a FastAPI task AI service using Google Gemini to detect tasks from comments and integrated Node cron schedulers for contact imports and lead/deal/task reminder emails.
- Worked with MySQL stored procedures, triggers, views, Swagger documentation, Helmet, rate limiting, JWT, and audit-log middleware.

Amzur AI Chat

Tech: React 18, TypeScript, Vite, Tailwind CSS, FastAPI, Python, PostgreSQL, Supabase, SQLAlchemy, LangChain, Gemini, LiteLLM, ChromaDB, Pandas, n8n

- Built a full-stack AI chat application with threaded conversations, streaming responses, conversation history, authentication, and interactive chat functionality.
- Implemented email/password and Google OAuth 2.0 authentication using secure HttpOnly JWT cookies and bcrypt.
- Developed multimodal chat supporting images, video, PDF, Excel, code, and formulas with server-side MIME validation.
- Implemented PDF RAG using ChromaDB and embeddings; integrated Gemini 2.5 Flash and Gemini Imagen 4.
- Built natural-language database querying that generates read-only SQL while blocking destructive SQL operations.
- Developed CSV, Excel, and Google Sheets analysis workflows using Pandas-based agents; integrated LiteLLM and n8n.

Internal Knowledge Base Navigator

Tech: Python, FastAPI, Streamlit, LangChain LCEL, ChromaDB, Gemini 2.5 Flash, LiteLLM, OpenAI Embeddings, pytest, HTTPX

- Built a conversational RAG application for answering employee questions from company policy documents.
- Designed intent classification, query rewriting, semantic search, grounded generation, and source-linking pipeline.
- Implemented ChromaDB vector search and text-embedding-3-large through LiteLLM.
- Developed typed FastAPI endpoints and implemented path-traversal protection and environment-based configuration.
- Developed 15 unit and integration tests using pytest with mocked LLM and vector-store calls.

EDUCATION

Bachelor of Technology (B.Tech) **Sep 2014 - May 2018**

Malla Reddy Institute of Technology

- Electronics and Communication Engineering (ECE)
- Final CGPA: 6.5

CERTIFICATIONS

- AI Agent Development Workshop – Frontlines Edutech | Build Your First AI Agent App Workshop
- AI Tools & ChatGPT Workshop – be10x | Completed July 2025
- AI Forge Capstone Project – Amzur | Completed AI learning program and capstone project, June 2026